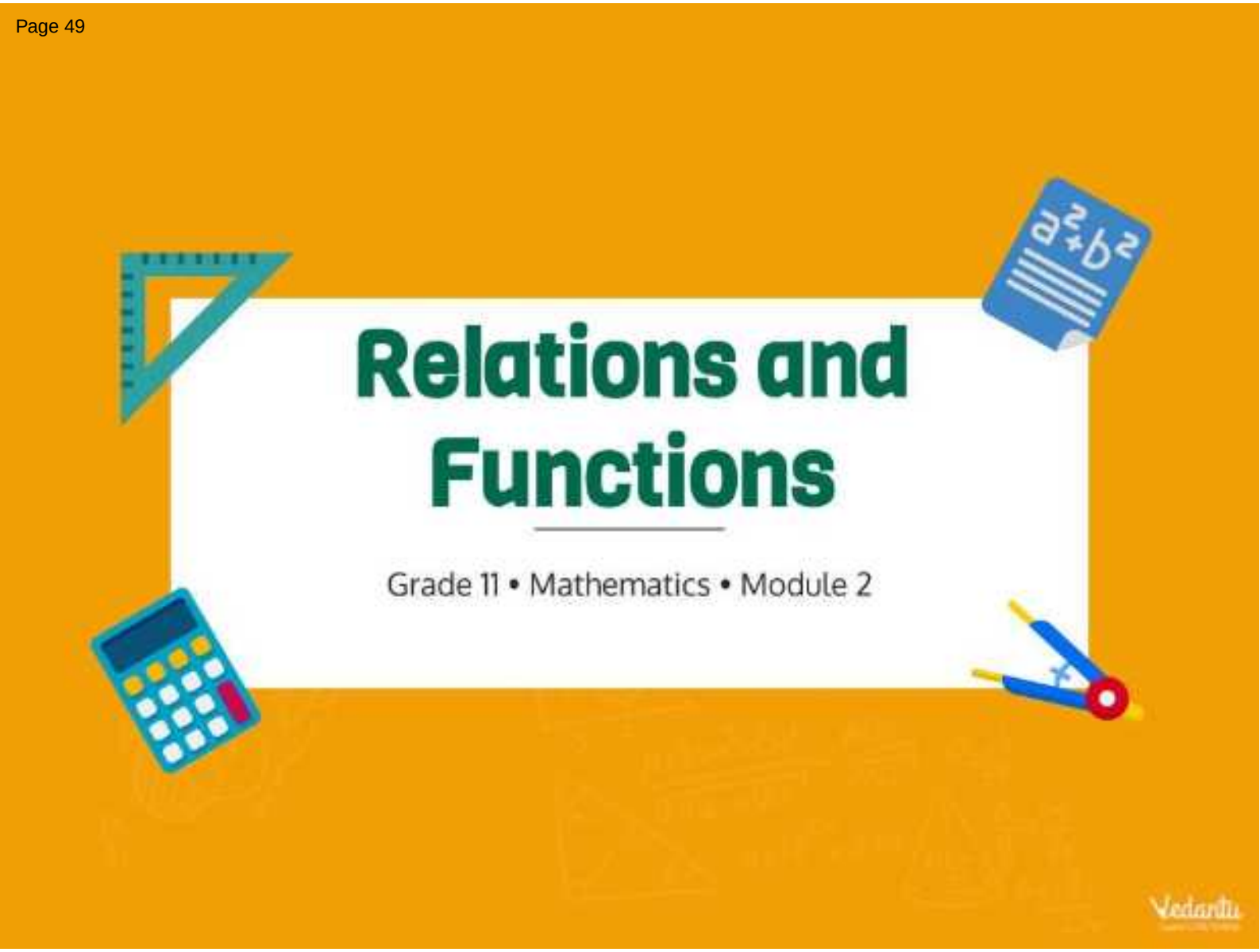

Relations and Functions
Grade 11 • Mathematics • Module 2



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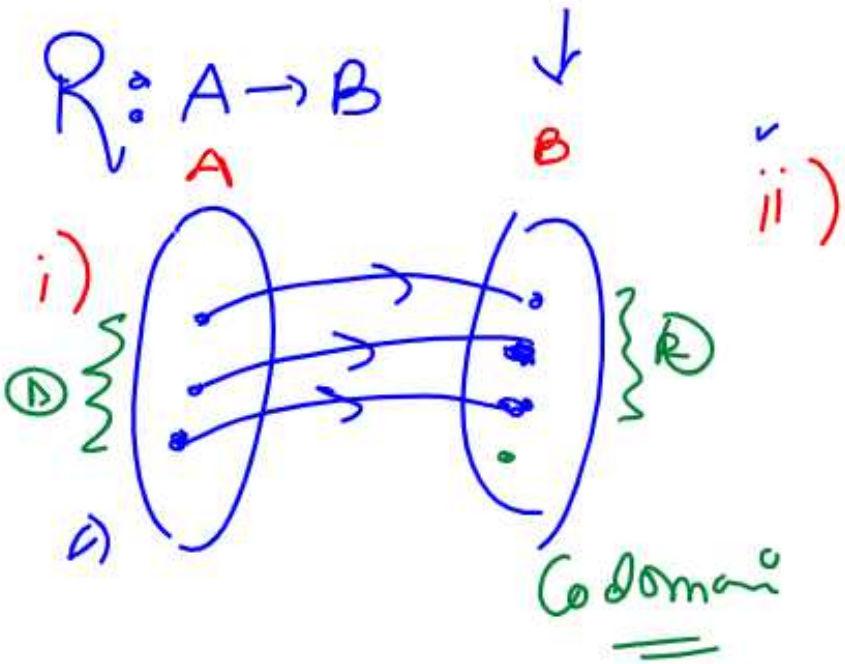
1 Sets

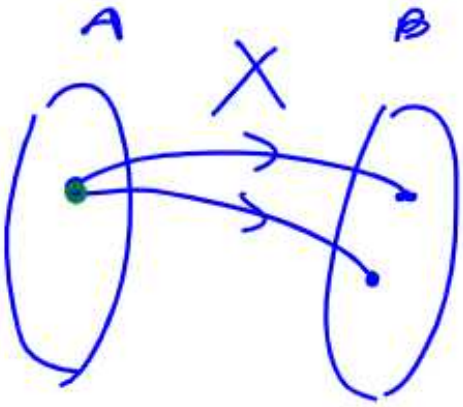
2 Relations

3 **Functions**

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FUNCTIONS





1) DOMAIN OF A FUNCTION

* $\sqrt{f(x)} \Rightarrow f(x) \geq 0$

* $\frac{1}{f(x)} \Rightarrow f(x) \neq 0$

* $\log_{g(x)} f(x) \Rightarrow f(x) > 0$, $g(x) > 0$, $g(x) \neq 1$

Let's Solve

Ex. Find the domain of the following.

(a) $y = \dfrac{1}{x^2 - 4}$

(b) $y = \sqrt{x^2 - 4}$

(c) $y = \dfrac{1}{\sqrt{x^2 - 4}}$

(d) $y = \log_{10}\left(x^2 - 1\right)$

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Let's Solve

Ex. Find the domain of the following.

(a) $y = \frac{1}{x^2 - 4}$

$x^2 - 4 \neq 0$

$x \neq \pm 2$

$x \in R - \{\pm 2\}$

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Let's Solve

Ex. Find the domain of the following.

(a) $y = \dfrac{1}{x^2 - 4}$

Solution

$$y = \frac{1}{x^2 - 4}$$

As denominator is polynomial so its domain is all real numbers except where it is zero.

$\therefore$ Domain $= \mathbb{R} - \{\pm 2\}$

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Let's Solve

Ex. Find the domain of the following.

(b) $y = \sqrt{x^2 - 4}$

i) $x^2 - 4 \geq 0$

$(x-2)(x+2) \geq 0$

$x \in (-\infty, -2] \cup [2, \infty)$

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Let's Solve

Ex. Find the domain of the following.

(b) $y = \sqrt{x^2 - 4}$

Solution

$y = \sqrt{x^2 - 4}$

For domain,

$x^2 - 4 \geq 0$

$x^2 \geq 4$

$x \leq -2$ or $x \geq 2$

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Let's Solve

Ex. Find the domain of the following.

(c) $y = \dfrac{1}{\sqrt{x^2 - 4}}$

i) $x^2 - 4 \geq 0$

ii) $x^2 - 4 \neq 0$

$\Rightarrow x^2 - 4 > 0$

$x \in (-\infty, -2) \cup (2, \infty)$

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Let's Solve

Ex. Find the domain of the following.

(c) $y = \dfrac{1}{\sqrt{x^2 - 4}}$

Solution

$$y = \frac{1}{\sqrt{x^2 - 4}}$$

For domain,

$x^2 - 4 > 0$

$x^2 > 4$

$x < -2$ or $x > 2$

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Let's Solve

Ex. Find the domain of the following.

(d) $y = \log_{10}\left(x^2 - 1\right)$

i) $x^2 - 1 > 0$ $\Rightarrow$ $x^2 - 1 > 0$

$(x-1)(x+1) > 0$

ii) $\cup$

$x \in (-\infty, -1) \cup (1, \infty)$

iii) $\cup$

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Let's Solve

Ex. Find the domain of the following.

(d) $y = \log_{10}\left(x^2 - 1\right)$

Solution

$y = \log_{10}\left(x^2 - 1\right)$

For domain,

$x^2 - 1 > 0$

$x^2 > 1$

$x < -1 \text{ or } x > 1$

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Quiz.



Quiz.

Chat Off

Q. The domain of the function $f(x)=\sqrt{x^2-3x+2}$ is

A $[1, 2]$

B $(1, 2)$

C $(-\infty, 1]\cup[2, \infty)$

D $(-\infty, 1)\cup(2, \infty)$

i) $x^2-3x+2 \geq 0$

$(x-1)(x-2) \geq 0$

$x \in (-\infty, 1] \cup [2, \infty)$



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Quiz.

Chat On

Q. The domain of the function $f(x)=\sqrt{x^2-3x+2}$ is

A $[1, 2]$

B $(1, 2)$

C $(-\infty, 1]\cup[2, \infty)$

D $(-\infty, 1)\cup(2, \infty)$

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Quiz.

Q. The domain of the function $f(x) = \sqrt{x^2 - 3x + 2}$ is

Solution

Clearly $f(x)$ is defined if $x^2 - 3x + 2 \geq 0$

$\Rightarrow (x-1)(x-2) \geq 0$

$\Rightarrow x \in (-\infty, 1] \cup [2, \infty)$

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Quiz.



Quiz.

Chat Off

Q. What is the domain of the function $f(x) = \log_2(2 - x^2)$?

A $x \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

B $x \in (-\infty, -\sqrt{2}] \cup [\sqrt{2}, \infty)$

C $x \in [-\sqrt{2}, \sqrt{2}]$

D $x \in (-\sqrt{2}, \sqrt{2})$



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Quiz.

Chat On

Q. What is the domain of the function $f(x) = \log_2(2 - x^2)$?

A $x \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

B $x \in (-\infty, -\sqrt{2}] \cup [\sqrt{2}, \infty)$

C $x \in [-\sqrt{2}, \sqrt{2}]$

D $x \in (-\sqrt{2}, \sqrt{2})$



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Quiz.

Q. What is the domain of the function $f(x) = \log_2(2 - x^2)$?

Solution

For domain,

$2 - x^2 > 0$

$\Rightarrow x^2 < 2$

$\Rightarrow x \in (-\sqrt{2}, \sqrt{2})$

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Let's Solve



Let's Solve

Ex. Find the domain of the following.

(a) $y = \log_x\left(4 - x^2\right)$

(b) $y = \sqrt{x^2 - 4} + \log_{10}(x - 5)$

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Let's Solve

Ex. Find the domain of the following.

(a) $y = \log_x\left(4 - x^2\right)$

i) $4 - x^2 > 0 \Rightarrow x^2 - 4 < 0$

$x \in (-2, 2)$

ii) $x > 0 \Rightarrow (x > 0)$

iii) $x \neq 1 \Rightarrow [x \neq 1]$

$x \in (0, 2) - \{1\}$

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Let's Solve

Ex. Find the domain of the following.

(a) $y = \log_x\left(4 - x^2\right)$

Solution

$y = \log_x\left(4 - x^2\right)$

For domain,

$x > 0$ & $\neq 1$ and $4 - x^2 > 0$

$x > 0$ & $\neq 1$ and $x^2 < 4$, that is $-2 < x < 2$

Taking common,

$x \in (0,1) \cup (1,2)$

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Let's Solve

Ex. Find the domain of the following.

(b) $y = \sqrt{x^2 - 4} + \log_{10}(x - 5)$

$D_1 \cap D_2$

$D_1 \Rightarrow x^2 - 4 \geq 0$

$(x-2)(x+2) \geq 0$

$x \in (-\infty, -2] \cup [2, \infty)$

$D_2 \quad x - 5 > 0$

$x \in (5, \infty)$

$\Rightarrow x \in (5, \infty)$

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Let's Solve

Ex. Find the domain of the following.

(b) $y=\sqrt{x^2-4}+\log_{10}(x-5)$

Solution

$y=\sqrt{x^2-4}+\log_{10}(x-5)$

For domain,

$x^2-4\geq 0$ and $x-5>0$

$x^2\geq 4$ and $x>5$

$x\leq -2$ or $x\geq 2$ and $x>5$

Thus, $x\in(5,\infty)$

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Quiz.



Quiz.

Chat Off

D1 D2

Q. What is the domain of the function $f(x)=\frac{1}{\sqrt{3-x}}+\log_x 3$?

A $x\in[0,3]-\{1\}$

B $x\in(0,3)-\{1\}$

C $x\in(0,3]-\{1\}$

D $x\in[0,3)-\{1\}$

i) $3-x>0 \Rightarrow x<3$

ii) $=$

iii) $x>0$

iv) $x\neq 1$

$\therefore x\in(0,3)-\{1\}$

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Quiz.

Chat On

Q. What is the domain of the function $f(x)=\frac{1}{\sqrt{3-x}}+\log_x 3$?

A $x \in [0, 3]-\{1\}$

B $x \in (0, 3)-\{1\}$

C $x \in (0, 3]-\{1\}$

D $x \in [0, 3)-\{1\}$



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Quiz.

Q. What is the domain of the function $f(x)=\frac{1}{\sqrt{3-x}}+\log_x 3$?

Solution

For domain,

$3-x>0$ and $x>0,\ x\neq 1$

$\Rightarrow x<3$ and $x>0,\ x\neq 1$

$\Rightarrow x\in(0,3)-\{1\}$



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Let's Solve



Let's Solve

H·W

Ex. Find the domain of $y = \sqrt{\dfrac{(x-1)(x-2)}{(9-4x^2)}}$.

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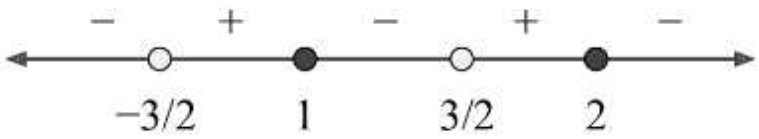
Let's Solve

Ex. Find the domain of $y=\sqrt{\dfrac{(x-1)(x-2)}{(9-4x^2)}}$.

Solution

For domain,

$$\frac{(x-1)(x-2)}{9-4x^2}\geq 0\ ;\ x\neq\pm\frac{3}{2}$$



$$\frac{(x-1)(x-2)}{(3-2x)(3+2x)}\geq 0$$

Thus, $x\in\left(-\frac{3}{2},1\right]\cup\left(\frac{3}{2},2\right]$

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Let's Solve

Ex. Solve $\sqrt{x^2-4}<3$ for x.

Let's Solve

Ex. Solve $\sqrt{x^2-4}<3$ for x.

Solution

$$\sqrt{x^2-4}<3 \qquad \ldots(1)$$

Here, $x^2-4\geq 0$

$\therefore x\in(-\infty,-2]\cup[2,\infty)$

Now squaring (1) on both sides

$$x^2-4<9$$

$$x^2<13$$

$$-\sqrt{13}<x<\sqrt{13}$$

Taking common, we get

$$(-\sqrt{13},-2]\cup[2,\sqrt{13})$$

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Summary.

3. Functions

   3.3 Classification of Functions

   3.4 Domain of a Function

Revise ⇒ Solve Question

(1) Find the domains of definition of the given functions.

(i) $y = \frac{1}{\sqrt{x^2 - 4x}}$ (ii) $y = \arcsin \frac{x}{4}$ (iii) $y = \arccos \frac{1-2x}{4}$

(iv) $y = \arcsin \sqrt{2x}$ (v) $y = \sqrt{x} + \sqrt[3]{\frac{1}{x-2}} - \log_{10}(2x-3)$

(vi) $y = \log_{10}[1 - \log_{10}(x^2 - 5x + 16)]$

[Ans. (i) $-\infty < x < 0$ and $4 < x < \infty$; (ii) $-4 \le x \le 4$; (iii) $-\frac{3}{2} \le x \le \frac{5}{2}$; (iv) $0 \le x \le \frac{1}{2}$

(v) $\frac{3}{2} < x < 2$ and $2 < x < \infty$; (vi) $2 < x < 3$.]

(2) Domain of the function $f(x) = \sqrt{1 - \sqrt{2 - \sqrt{3 - x}}}$ is

(A) [0, 2] (B) [–1, 1] (C*) [–1, 2] (D) [1, 2]

(i) $f(x) = \sqrt{\sin x} + \sqrt{16 - x^2}$ (ii) $f(x) = \sqrt{\log_2\left(\frac{5x - x^2}{4}\right)}$ or $\sqrt{\log_{\frac{1}{2}} \frac{5x - x^2}{4}}$

(iii) $f(x) = \cos^{-1}\left(\frac{2-|x|}{4}\right) + (\ln(3 - x))^{-1}$ [Ans. $[-6, 2] \cup (2, 3)$;

(iv) $f(x) = \frac{1}{\sqrt{|x| - x}}$ $(x < 0)$; $h(x) = \sqrt{\frac{x-2}{x+2}} + \sqrt{\frac{1-x}{1+x}}$

$g(x) = \frac{1}{\sqrt{x - |x|}}$ [Ans: ϕ] ; $l(x) = \log_{2[x]-3}(x^2 - 5x + 13)$ [Ans: ϕ]

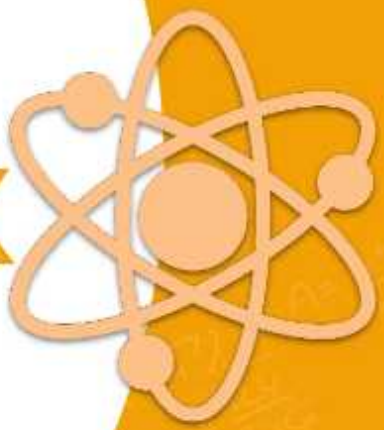
(v) $f(x) = \sqrt{\log_x(\cos 2\pi x)}$ [Ans. $(0, \frac{1}{4}) \cup (\frac{3}{4}, 1) \cup \{2, 3, 4, ... \infty)$] **[T/S]**

(viii) $f(x) = \left(\log_{\frac{x-2}{x+3}} 2\right) + \sqrt{9 - x^2}$ {Ans. (2, 3] }

(vi) $f(x) = \frac{1}{\sqrt{\sin(\cos x)}} + \sin^{-1}\left(\frac{2x}{\pi}\right) + \frac{1}{\{-x\}} + \frac{1}{\ln\left(1 - \left[\tan\frac{x}{2}\right] - \left[-\tan\frac{x}{2}\right]\right)}$



Feedback



Thanks.

